

OLTENITA, Romania

WMO: 154750

Lat: 44.0747N

Lon: 26.6372E

Elev: 16

StdP: 101.14

Time Zone: 2.00 (EUE)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.2	-8.8	-13.2	1.2	-10.9	-10.5	1.5	-7.5	10.8	-5.8	9.8	-0.9	1.8	250	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.6	33.8	22.6	32.5	22.0	31.2	21.4	24.3	31.2	23.5	30.0	22.7	28.9	2.6	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.4	17.1	26.7	21.6	16.2	26.0	20.7	15.4	25.3	73.1	31.2	70.0	29.9	67.0	29.0	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	7.4	6.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.4	-1.2	3.8	7.5	11.7	17.6	22.2	23.6	23.7	19.1	11.8	6.2	2.2		
	DBStd	9.29	5.05	4.90	4.67	4.12	3.25	2.83	2.43	2.63	3.69	4.21	4.54	3.81		
	HDD10.0	1063	346	178	103	30	1	0	0	0	31	131	244			
	HDD18.3	2709	604	407	336	199	53	4	0	1	36	204	363	501		
	CDD10.0	1940	0	5	26	82	236	364	422	425	274	88	17	1		
	CDD18.3	544	0	0	0	1	29	119	164	168	59	3	0	0		
	CDH23.3	5508	0	0	4	56	314	1087	1546	1838	617	45	2	0		
	CDH26.7	2184	0	0	0	9	67	410	629	847	214	8	0	0		
(14) Wind	WSAvg	2.3	2.4	2.5	2.5	2.6	2.5	2.1	2.0	2.0	2.0	2.0	2.0	2.6		
(15) Precipitation	PrecAvg	595	44	29	41	48	60	67	62	38	55	50	43	50		
	PrecMax	800	114	93	99	128	208	172	144	123	164	127	95	148		
	PrecMin	364	7	1	2	3	13	17	6	3	1	2	0	0		
	PrecStd	128	27	20	23	25	44	35	36	31	41	32	27	38		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	20.3	23.2	28.0	31.3	35.1	35.6	35.5	33.2	28.4	22.5	15.0		
		MCWB	9.0	14.2	15.6	17.6	22.0	24.2	24.5	23.1	20.8	18.8	15.7	11.2		
	2%	DB	9.8	15.7	20.5	24.7	28.6	32.5	33.3	33.9	31.2	24.4	18.4	12.1		
		MCWB	7.2	10.9	13.4	16.2	19.6	22.8	22.6	22.3	20.0	16.9	14.2	8.9		
	5%	DB	7.4	13.3	18.0	22.1	26.9	30.8	31.9	32.6	29.2	21.8	15.7	9.5		
		MCWB	5.7	9.4	12.0	14.7	18.8	21.6	22.1	21.7	19.3	15.7	12.6	7.3		
	10%	DB	5.1	11.0	15.4	19.9	24.9	29.1	30.1	31.3	27.1	19.4	13.0	7.1		
		MCWB	4.2	8.0	10.8	13.7	17.9	21.4	21.2	21.4	19.0	14.5	11.2	5.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	14.0	17.3	18.3	22.8	26.1	25.6	25.0	23.7	19.6	16.2	11.7		
		MCDB	12.4	19.6	22.4	26.4	28.8	33.4	34.0	31.9	30.5	25.7	21.4	15.1		
	2%	WB	7.5	11.4	14.2	17.0	20.9	24.0	24.1	24.1	22.0	17.6	14.7	9.3		
		MCDB	9.6	15.1	19.0	23.9	26.7	30.6	31.2	30.8	27.5	22.8	18.6	11.8		
	5%	WB	5.8	9.8	12.6	15.7	19.7	22.9	23.2	23.3	20.7	16.3	13.1	7.7		
		MCDB	7.0	12.7	17.3	21.6	25.0	29.2	29.9	29.6	26.0	20.9	15.4	9.5		
	10%	WB	4.3	8.3	11.1	14.4	18.6	22.0	22.3	22.6	19.7	15.2	11.6	6.0		
		MCDB	5.1	10.8	15.2	18.9	23.7	27.9	28.2	28.6	25.0	18.8	13.3	7.2		
(35) Mean Daily Temperature Range	MDBR		6.8	8.8	10.4	11.7	11.5	11.4	11.7	13.6	12.7	11.2	7.1	6.3		
	5% DB	MCDBR	9.8	13.6	15.6	16.7	14.6	14.4	14.3	15.6	16.3	16.2	12.0	10.6		
		MCWBR	7.3	8.8	9.0	8.9	6.7	5.9	5.8	6.5	8.9	7.1	7.6			
	5% WB	MCDBR	9.2	12.4	15.3	15.7	12.4	13.5	12.6	13.4	13.7	14.5	10.5	10.5		
		MCWBR	7.1	8.5	9.1	8.5	6.3	6.2	5.6	5.5	6.0	8.3	6.4	7.5		
(40) Clear-Sky Solar Irradiance	taub		0.352	0.385	0.421	0.462	0.435	0.443	0.453	0.467	0.434	0.405	0.377	0.355		
	taud		2.262	2.185	2.141	2.047	2.203	2.215	2.196	2.151	2.207	2.264	2.291	2.276		
	Ebn at Noon		743	783	805	803	840	832	817	790	783	753	709	697		
	Edh at Noon		88	113	134	160	141	139	141	142	124	103	86	80		
(44) All-Sky Solar Radiation	RadAvg		1.42	2.22	3.38	4.68	5.86	6.36	6.46	5.79	4.20	2.72	1.54	1.18		
	RadStd		0.22	0.32	0.34	0.50	0.43	0.46	0.43	0.31	0.32	0.30	0.20	0.16		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	-119	+70	+41

Nomenclature: See separate page